Comparative Analysis of Deep Learning Architectures for Bias-Aware Toxicity Classification

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***Abstract*—Unintended bias in automated toxicity detection — where models falsely associate identity-related terms with toxicity — remains a significant challenge in NLP. This paper presents a comparative analysis of two deep learning architectures: Parallel Hybrid with Multi-Head Self-Attention (PH-MSA), which combines multi-scale CNNs and BiGRU for rich contextual feature extraction, and TinyBERT with Fairness-Regularized Distillation (TFRD), which employs a teacher-student framework with demographic fairness regularization. Experiments on the Jigsaw Unintended Bias dataset demonstrate that PH-MSA achieves higher accuracy (95%) and precision (0.78), while TFRD achieves a superior macro-F1 score (0.80) with more balanced performance across identity groups, confirming that fairness-aware training objectives are essential for building reliable and equitable toxicity detection systems.**

***Index Terms*—Toxicity Detection, Deep Learning, Unintended Bias, NLP, Fairness in AI**

I. Introduction

Online communities, while revolutionizing digital collaboration, have also increased the wide spread of toxic content, which includes hate speech and identity-based harassment, which are significant threats to vulnerable groups and mental well-being [1]. Although DL models employing attention mechanisms are more accurate than traditional rule-based methods [2], two significant issues remain which are, handling contextual dependencies and reducing demographic biases, where models incorrectly correlate identity-based words such as race and religion with toxicity [3], [4]. However, most methods are optimized for classification metrics while ignoring demographic fairness, which is a critical limitation in building fair toxicity detection models.

In this study, we compare two models: PH-MSA and TFRD, which are intended to address contextual representation and demographic biases in toxicity classification and are assessed using the Jigsaw Unintended Bias dataset. Although PH-MSA indirectly deals with demographic biases using its parallel architecture, TFRD directly deals with demographic biases using fairness-regularized knowledge distillation.

The key contributions of this study are as follows: (i) a new PH-MSA architecture that integrates multi-scale CNNs and BiGRU along with multi-head self-attention for more comprehensive contextual feature learning; (ii) a framework for fairness-regularized distillation, namely TFRD, that directly seeks to minimize error disparity on demographic groups during training; and (iii) a comparative analysis of both models on the Jigsaw Unintended Bias dataset, considering a variety of evaluation metrics.

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II. Related Works

There have been significant advancements in the field of toxicity and hate speech classification, especially using models that focus on semantic feature extraction. The use of transformer-based models is prominent due to their ability to handle contextual dependencies. Mazari et al. proposed a BERT-based ensemble for multi-aspect hate speech classification, resulting in greater robustness for hate speech from diverse dimensions [5].

Hybrid models have also played a significant role, especially in modeling global and local structures. Wang et al. proposed a dual BiGRU and CNN architecture for sentiment classification, incorporating global and local attention. The proposed architecture showed significant improvement for sentiment classification [6]. Soni et al. proposed TextConvoNet, proving that CNNs can hold their own against transformer-based models for text classification tasks [7]. Nelatoori and Kommanti proposed a multi-task learning architecture for toxic comment classification and rationale extraction, resulting in transparency for decision-making processes [8]. Almerekhi et al. proposed PROVOKE, proving that most cases of toxicity occur due to conversational patterns and not statements [9]. But the problem of demographic bias is still untouched, despite achieving significant performance for text classification tasks.

Subsequent studies have sought to employ contrastive learning, multimodal approaches, and network-aware modeling to further advance robustness. For example, Lu et al. (2023) presented a dual contrastive learning paradigm to mitigate dependence on misleading lexical patterns [10]. Nivedita and Banu (2025) used a hybrid contrastive graph neural network, incorporating both text and user interaction, to detect coordinated hate speech. Saddozai et al. (2025) extended robustness to multimodal and multilingual settings by jointly processing text and image data. Hashir et al. (2025) presented TARGE, a hate speech detection system using LLMs for explainable approaches, but there remains concern about bias from the pre-trained models used [13].

Fairness and bias issues have become major concerns as these approaches near deployment. For instance, Shah and Sureja (2025) investigated the implications of bias from deep learning models on society [14]. Jotheeswaran et al. (2025) showed that toxic comment classifiers can be used as part of automated moderation processes [15]. These studies suggest that there is a significant gap between performance and fairness, and this is what this comparative analysis aims to bridge.

III. Methodology

The overall flow of the methodological pipeline followed for this research work is presented in Fig. 1. The pipeline depicts the preparation, training, and evaluation process involved that provide an overview of how raw input data is processed to develop the performance results. All steps in this pipeline are discussed in detail in the next few sections.

A. Dataset Used

We test our models on the Jigsaw Unintended Bias in Toxicity Classification dataset, a large-scale benchmark dataset that aims to test the toxicity classification as well as the unintended demographic bias in the models. The dataset contains a set of 2 million English-language online commentaries, and each of these has a continuous measure of toxicity based on the ratings given by the annotators.

Besides the toxicity label, the fine-grained labels included in the dataset are insult, identity attacks, severe toxicity, and binary identity indicators based on gender, religion, ethnicity, sexual orientation, and disability. Due to the huge imbalance in the total data, a strategically chosen dataset of 100,000 data points is considered, which maintains the original distribution of the labels. The 100,000 data points are randomly chosen from the total 1.8M comment training set using a stratified sampling technique based on the binary toxicity label (scikit-learn's train_test_split function, stratify=y, random_state=42). The 100,000 data points have the same 8% toxic and 92% non-toxic class distribution as the full training set.

The PH-MSA model was trained on an 80/20 split of the data, and the TFRD model was trained on a 90/10 split of the data. Both the splits were stratified. Over sampling techniques were not applied. Class imbalance is addressed through macro-averaged F1 and for TFRD, the fairness regularization loss.

B. Proposed Frameworks

The raw comments from the Jigsaw Unintended Bias dataset were preprocessed following a standardized process that included text normalization, removal of URLs, HTML tags, and special characters; handling of missing values; and binarization of continuous toxicity scores based on a threshold 0.5. Identity features were retained to enable bias analysis; irrelevant features were discarded. Two different deep learning architectures were trained on the preprocessed data: PH-MSA and TFRD. Each architecture had a unique objective: to maximize contextual accuracy along with awareness of identities and to achieve fairness-conscious balanced learning. This approach ensures that results are not dependent on a specific modeling approach. Both architectures are discussed in detail in the following subsections.

1) TinyBERT with Fairness-Regularized Distillation (TFRD): The TFRD model is a combination of knowledge distillation and fairness-aware regularization under a transformer architecture for toxicity classification, as shown in Figure 2. A pre-trained BERT-base is used as the teacher model, while the student is a compact TinyBERT. During training, the student learns from both label and teacher predictions by leveraging transformer guidance. A fairness-aware regularization is added to reduce the gap between false positive rates of identity and non-identity groups. As shown in Algorithm 1, the optimization objective is defined by the weighted sum of the task loss, distillation loss, and fairness loss to enable the model to maintain competitive performance while minimizing demographic bias.

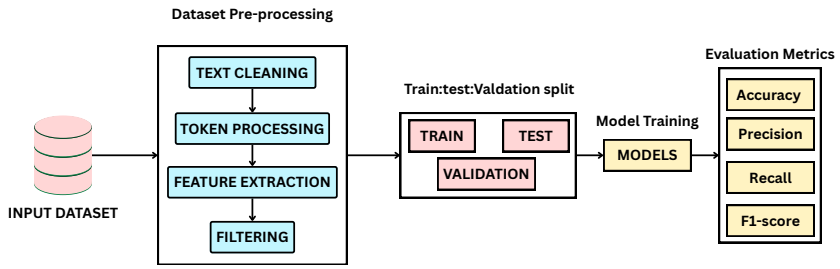

Fig. 1: Methodology of the study.

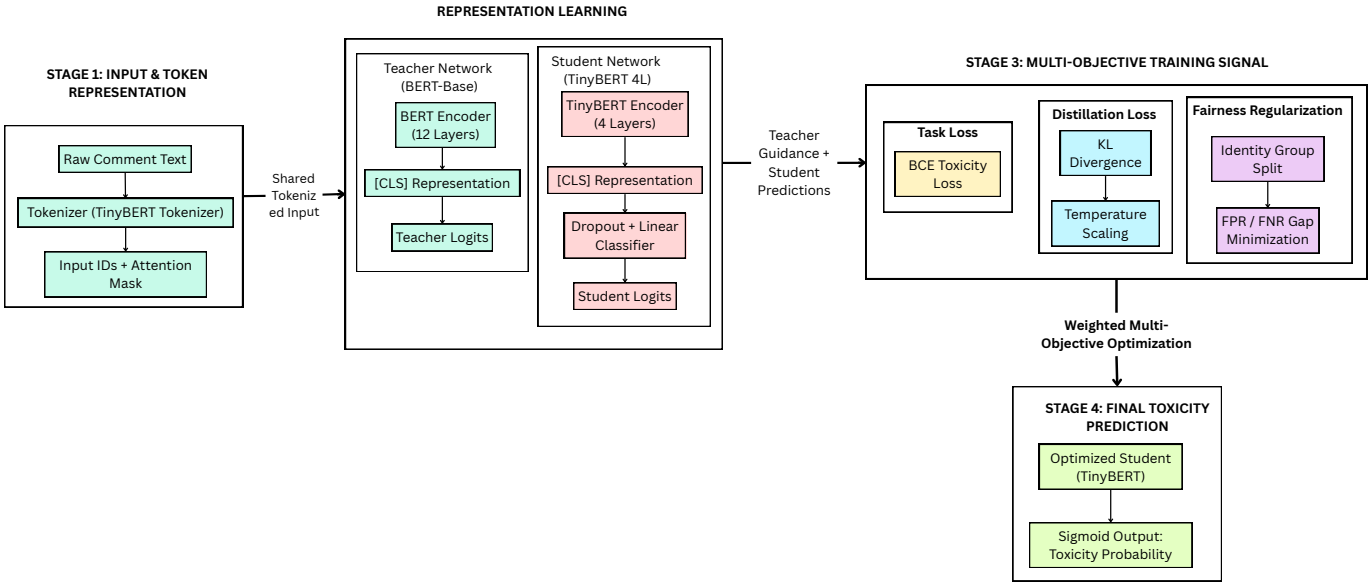

Fig. 2: TFRD architecture Diagram

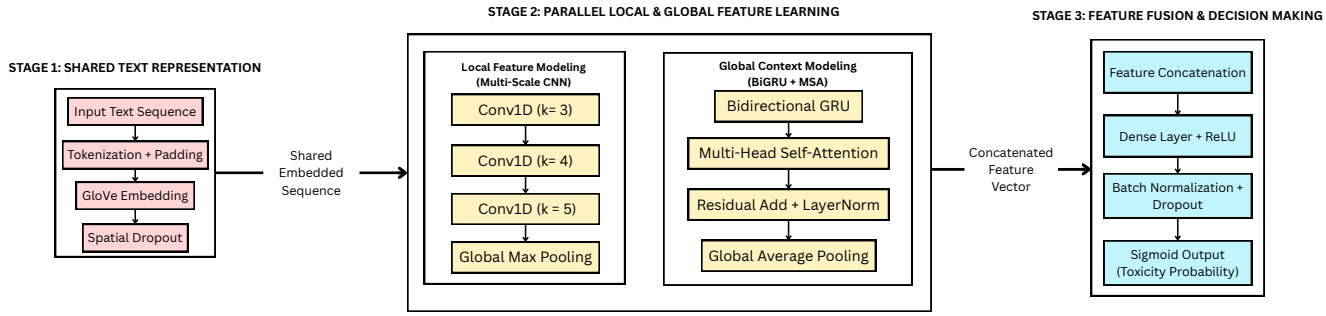

Fig. 3: PH-MSA architecture Diagram

2) Parallel Hybrid with Multi-Head Self-Attention (PH-MSA): The proposed architecture of the PH-MSA model is presented in Figure 3, where the local lexical features and the long-range contextual features are both well exploited for the toxic comment classification task. In comparison to the previous hybrid models like Wang et al. (2024) [6], where the CNN and RNN branches are sequentially operated with independent attention mechanisms, the PH-MSA architecture uses both branches of the CNN and BiGRU in parallel. The inputs of the PH-MSA are GloVe word embeddings. The GloVe word embeddings are fed into the two branches of the PH-MSA architecture. In the first branch of the PH-MSA architecture, the GloVe word embeddings are operated by the multi-scale CNNs with the kernel size of 3, 4, and 5 to learn the local features of the toxic language. In the second branch of the PH-MSA architecture, the GloVe word embeddings are operated by the Bidirectional GRU and the Multi-Head Self-Attention mechanism to learn the long-range contextual features. The outputs of the two branches are concatenated and operated by the fully connected layer. The training process of

Algorithm 1 TFRD: TinyBERT with Fairness-Regularized Distillation

Require: Preprocessed dataset $D = \{(x_i, y_i, I_i)\}$
Require: Teacher model T (BERT-base), Student model S (TinyBERT)
Require: Hyperparameters $\alpha_{task}, \alpha_{distill}, \alpha_{fair}$
Freeze parameters of teacher model T
for each training epoch **do**
  for each mini-batch (x, y, I) **do**
    Compute teacher logits $z_T = T(x)$
    Compute student logits $z_S = S(x)$
    Compute task loss L_{task} using binary cross-entropy
    Compute distillation loss $L_{distill}$ between z_S and z_T
    Compute fairness loss L_{fair} using identity-based error disparity
    Compute total loss:

$$L = \alpha_{task} L_{task} + \alpha_{distill} L_{distill} + \alpha_{fair} L_{fair}$$

    Update student parameters using backpropagation
  end for
end for
Select model with best validation AUC

the PH-MSA is presented in Algorithm 2.

C. Comparative Experimental Methodology

Both models were trained and validated on the same preprocessed 100,000-sample Jigsaw subset under reproducible conditions (random seed = 42), ensuring a systematic and fair basis for comparison. Full hyperparameter configurations are provided in Table I. Evaluation was conducted on the same validation partition using identical metrics. Neither model uses early stopping; PH-MSA applies a ReduceLROnPlateau scheduler while TFRD uses a linear warmup schedule, in both cases, the checkpoint with the best validation AUC was retained. Training hyperparameters differ out of architectural necessity: transformer fine-tuning demands a smaller learning rate and fewer epochs to avoid catastrophic forgetting of pre-trained weights, whereas PH-MSA benefits from a higher learning rate and longer training.

For the comparative evaluation of the proposed models, the metrics used are Accuracy, F1-score, and Area Under the ROC Curve (AUC). In addition to these metrics, the proposed approach has also been evaluated using a fairness-aware approach by analyzing the disparity in the prediction of the models for different groups of identities and non-identities.

IV. Results

A. Performance of Proposed Models

Table II presents the performance of both models evaluated on precision, recall, F1-score, and accuracy. Due to large class-imbalance, we emphasize performance on the minority toxic class. Table III shows how both proposed models compare with existing techniques, emphasizing their fairness aware design.

Algorithm 2 PH-MSA: Parallel Hybrid with Multi-Head Self-Attention

Require: Preprocessed dataset $D = \{(x_i, y_i)\}$
Require: GloVe embedding matrix E
Require: Hyperparameters for CNN, BiGRU, Attention, and Dense layers
for each training epoch **do**
  for each mini-batch (x, y) **do**
    Convert text to token sequences
    Map tokens to embeddings using pre-trained GloVe
    Apply spatial dropout
    Branch 1: Multi-Scale CNN
    for each kernel size k in {3,4,5} **do**
      Apply 1D convolution with kernel size k
      Apply global max pooling
    end for
    Concatenate pooled CNN features
    Branch 2: BiGRU + Multi-Head Attention
    Pass embeddings through Bidirectional GRU
    Apply Multi-Head Self-Attention
    Add residual connection and layer normalization
    Apply global average pooling
    Feature Fusion
    Concatenate CNN and BiGRU-Attention features
    Pass through dense layers
    Compute output probability using sigmoid activation
    Compute binary cross-entropy loss
    Update model parameters via backpropagation
  end for
end for
Select model with best validation AUC

Figure 4 shows that in the TFRD confusion matrix, there are 4,457 true negatives, 253 true positives, 148 false positives, and 142 false negatives, with precision and recall values balanced at 0.64. The ROC curve of TFRD with AUC = 0.9513 climbs steeply and closely follows the top left corner of the plot, suggesting high separability with low false positive rates. The PR curve of TFRD with AP = 0.7345 keeps high precision with high recall values and then declines gradually. The PH-MSA confusion matrix in Figure 5 indicates 18,173 true negatives and 770 true positives, with a relatively high 829 false negatives, consistent with its relatively low recall of 0.49. The ROC curve of PH-MSA with AUC = 0.9427 climbs steeply, though slightly lower than TFRD. The PR curve of PH-MSA with AP = 0.6948 declines rapidly after recall reaches 0.4, suggesting difficulty in identifying toxic samples. In imbalanced cases, PR curves are generally considered more meaningful than ROC curves. The PR-AUC of TFRD is indeed higher, confirming TFRD's stronger performance in detecting the minority class.

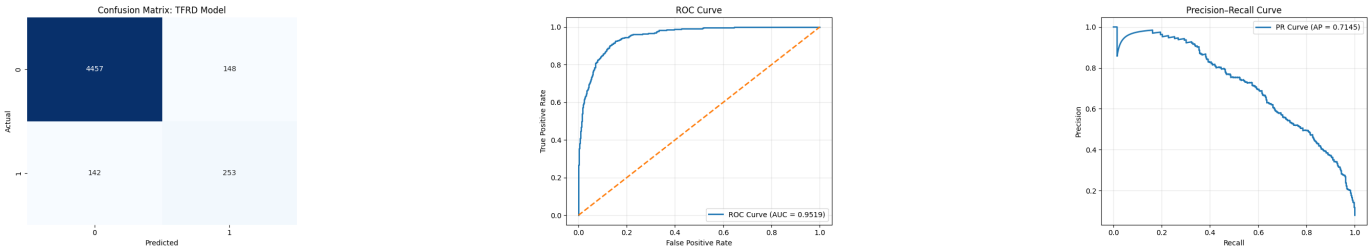

Fig. 4: TFRD Model Performance: (left) Confusion Matrix, (center) ROC Curve, (right) PR Curve

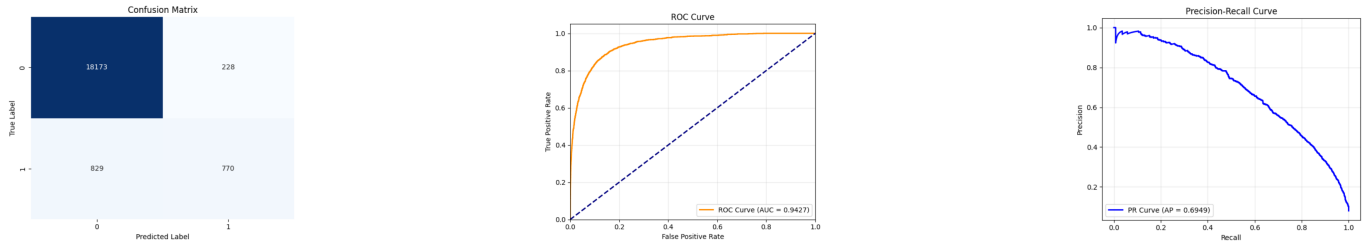

Fig. 5: PH-MSA Model Performance: (left) Confusion Matrix, (center) ROC Curve, (right) PR Curve

TABLE I: Hyperparameter Configurations for PH-MSA and TFRD

Hyperparameter	**PH-MSA**	**TFRD (student)**
Embedding	GloVe 300d (frozen)	WordPiece, max_len=256
Vocab / seq len	50k tokens / 300	30,522 / 256
Architecture	CNN(k=3,4,5) + Bi-GRU + MHA	TinyBERT 4L (hidden=312)
Teacher	—	BERT-base-uncased (frozen)
Batch size	128	32
Learning rate	0.001 (Adam)	2e-5 (AdamW)
Epochs	15	3
Dropout	0.2 spatial / 0.3 dense	0.3
LR scheduler	ReduceLROnPlateau (p=2)	Linear warmup (r=0.1)
Distill temp / weights	—	T=4.0 / α=0.3/0.5/0.2
Fairness λ	—	0.1
Train / Val split	80% / 20% (stratified)	90% / 10% (stratified)
Random seed	42	42

TABLE II: Performance Comparison of Proposed Models

Model	**Accuracy**	**Precision**	**Recall**	**F1-score**	**Macro F1**
TFRD	0.94	0.64	0.64	**0.64**	**0.80**
PH-MSA	**0.95**	**0.78**	0.49	0.60	0.79

TABLE III: Qualitative Comparison with Existing Methods

Method	**Architecture**	**Task**	**Fairness-Aware**	**Bias Metrics**
Mazari et al. [5]	BERT Ensemble	Hate Speech	No	No
Wang et al. [6]	BiGRU-CNN	Sentiment	No	No
Soni et al. [7]	CNN	Text Class.	No	No
Nelatoori et al. [8]	Multi-task CNN	Toxicity	No	No
Lu et al. [10]	Contrastive	Hate Speech	No	No
Hashir et al. [13]	LLM	Hate Speech	Partial	No
TFRD (ours)	TinyBERT+KD	Toxicity	**Yes**	**Yes**
PH-MSA (ours)	CNN+BiGRU +MHA	Toxicity	Partial	**Yes**

PH-MSA obtains the highest accuracy rate (95%), primarily due to high accuracy on the majority class "non-toxic." Its high precision value of 0.78 ensures a very low number of false positives. However, PH-MSA's low recall value of 0.49 means a very conservative approach to detection, as evidenced by its F1-score on the toxic class, which is 0.60. TFRD, on the other hand, obtains balanced precision and recall values of 0.64 and a macro-F1 of 0.80, indicating that fairness in distillation improves sensitivity to minority classes even when accuracy is slightly lower (94%).

B. Comparative Analysis of PH-MSA and TFRD

PH-MSA and TFRD differ significantly in the approach to bias-aware toxicity classification. PH-MSA's precision focused and hybrid architecture achieves higher precision (0.78) and accuracy (95%) which make the model more suitable for scenarios where false positives are critical, but its poor recall rate (0.49) makes it unsuitable for scenarios where the absence of toxic content is essential. On the other hand, the fairness-regularized distillation approach of TFRD is more suitable for scenarios where class equilibrium is a priority, as demonstrated by the higher macro-F1 score of 0.80. With regard to the computational cost of the two models, PH-MSA ($\sim$16.5M parameters) is lighter and faster at inference than TFRD's TinyBERT student ($\sim$14.5M parameters), with the BERT-base teacher ($\sim$110M) used only during training. The BERT-base of

approximately 110 million parameters is used for the teacher but not the student. The choice between the two models depends on the deployment scenario's priority: conservative filtering or fairness-aware detection.

C. Fairness Evaluation

For quantification of bias mitigation, both models have been evaluated on false positive rate (FPR) disparity, false negative rate (FNR) disparity, and Equalized Odds Gap for identity-mentioning and non-identity-mentioning subgroups. The results have been presented in Table IV.

TABLE IV: Fairness Metrics Across Identity Subgroups

Metric	PH-MSA	TFRD
FPR — non-identity	0.011	0.030
FPR — identity	0.025	0.066
FPR Disparity	0.014	0.037
FNR — non-identity	0.502	0.330
FNR — identity	0.604	0.480
FNR Disparity	0.102	0.150
Equalized Odds Gap	**0.058**	0.093

In Table IV, we have presented the fairness metrics for both models on identity-mentioning and non-identity-mentioning subgroups. From the results, it is evident that the proposed model PH-MSA has achieved lower Equalized Odds Gap (0.058) compared to TFRD (0.093). The lower FPR disparity of the proposed model (0.014) compared to TFRD (0.037) implies that the proposed model is more capable of handling false positives for identity-mentioning text. On the other hand, TFRD has achieved higher FPR disparity despite its explicit fairness regularization. The increase in FPR disparity of TFRD implies that the distillation objective and fairness regularization may have had some competition during training. We also observe FNR disparity across identity groups for both models, which is inevitable for detecting toxic comments that reference identity terms in the Jigsaw dataset. Overall, this demonstrates that architectural hybridization and fairness-regularized distillation are complementary but limited techniques for addressing demographic bias in ML models and warrants further research into minority class-aware objectives.

V. Conclusion and Future Directions

This paper compared PH-MSA and TFRD for their performance in bias-aware toxicity classification tasks. PH-MSA was found to have high precision (0.78) and accuracy (95%), although its low recall (0.49) was a drawback for PH-MSA. In contrast, the fairness-regularized distillation performed better than PH-MSA for macro-F1 scores (0.80), indicating that fairness-conscious training methods can be an effective alternative for designing new architectures for mitigating identity bias.

Future work may include the extension of the evaluation of the PH-MSA and TFRD models by adding new benchmark datasets like Civil Comments and HatEval. Moreover, the models would be beneficial if they take into consideration the various identity categories of the individuals being targeted by hate speech. In most cases of hate speech, the targets are often individuals from various overlapping categories of marginalized identity. Furthermore, the incorporation of Explainable AI would be beneficial for the deployment of the models. Explainable AI would be useful for the interpretation of the results of the models.

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